DATA ANALYSIS USING SQL

Situation: Analyze a large amount of historical data related to a **Major League Baseball (MLB).**

You have access to decades worth of data including player statistics like schools attended, salaries, teams played for, height & weight and more

Task: The task is to use the advanced SQL querying techniques to track how player statistics have changes over time and across different teams in the league

Action & Result:

PART I: SCHOOL ANALYSIS

1. View the schools and school details tables

`SELECT * FROM schools;`

Result Grid | Filter Rows:

playerID	schoolID	yearID
aardsda01	pennst	2001
aardsda01	rice	2002
aardsda01	rice	2003
abadan01	gamiddl	1992
abadan01	gamiddl	1993
abbeybe01	vermont	1889
abbeybe01	vermont	1890
abbeybe01	vermont	1891
abbeybe01	vermont	1892

`SELECT * FROM school_details;`

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell C

schoolID	name_full	city	state	country
abilchrist	Abilene Christian University	Abilene	TX	USA
adelphi	Adelphi University	Garden City	NY	USA
adrianmi	Adrian College	Adrian	MI	USA
akron	University of Akron	Akron	OH	USA
alabama	University of Alabama	Tuscaloosa	AL	USA
alabamaam	Alabama A&M University	Normal	AL	USA
alabamast	Alabama State University	Montgomery	AL	USA
albanyst	Albany State University	Albany	GA	USA
albertsnid	Albertson College	Caldwell	ID	USA

2. In each decade, how many schools were there that produced players?

```sql
SELECT   FLOOR(yearID / 10)*10 AS decade,
         COUNT(DISTINCT schoolID) AS num_schools
FROM     schools
GROUP BY decade
ORDER BY decade;
```

Result Grid | Filter Rows: | Export:

decade	num_schools
1860	2
1870	14
1880	34
1890	89
1900	148
1910	178
1920	196
1930	162
1940	142
1950	176
1960	301
1970	427
1980	473
1990	494

3. What are the names of the top 5 schools that produced the most players?

```
SELECT  sd. name_full, COUNT(DISTINCT sc.playerID) AS num_players
FROM    schools as sc LEFT JOIN school_details as sd
        on sc.schoolID = sd.schoolID
GROUP BY sc.schoolID
ORDER BY num_players DESC
LIMIT 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

name_full	num_players
University of Texas at Austin	107
University of Southern California	105
Arizona State University	101
Stanford University	86
University of Michigan	76

4. For each decade, what were the names of the top 3 schools that produced the most players?

```
-- For each decade, what were the names of the top 3 schools that produced the most players
with ds as (SELECT  floor(sc.yearID /10) *10 as decade, sd. name_full,
            COUNT(DISTINCT sc.playerID) AS num_players
            FROM    schools as sc LEFT JOIN school_details as sd
                    on sc.schoolID = sd.schoolID
            GROUP BY decade, sd.name_full),
    rn as (select decade, name_full, num_players,
    dense_rank() over(partition by decade order by num_players desc) as row_num
    from ds)
SELECT decade, name_full, num_players FROM rn
WHERE row_num <= 3
order by decade desc, num_players;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

decade	name_full	num_players
2010	Georgia Institute of Technology	3
2010	University of South Carolina	3
2010	University of Texas at Austin	4
2010	University of Florida	5
2000	Louisiana State University	20
2000	Stanford University	22
2000	California State University Long...	23
2000	Arizona State University	23
1990	Louisiana State University	22
1990	University of Southern California	23
1990	Stanford University	25
1980	University of California, Los An...	22
1980	Arizona State University	23

PART II: SALARY ANALYSIS

1. View the salaries table

```sql
-- View the salaries table
SELECT * FROM salaries;
```

Result Grid | Filter Rows: | Edit:

yearID	teamID	lgID	playerID	salary
1985	ATL	NL	barkele01	870000
1985	ATL	NL	bedrost01	550000
1985	ATL	NL	benedbr01	545000
1985	ATL	NL	campri01	633333
1985	ATL	NL	ceronri01	625000
1985	ATL	NL	chambch01	800000
1985	ATL	NL	dedmoje01	150000
1985	ATL	NL	forstte01	483333
1985	ATL	NL	garbege01	772000
1985	ATL	NL	harpete01	250000
1985	ATL	NL	hornebo01	1500000
1985	ATL	NL	hubbagl01	455000
1985	ATL	NL	mahleri01	407500

salaries 2

2. Return the top 20% of teams in terms of average annual spending

```sql
-- Return the top 20% of teams in terms of average annual spending
WITH ts AS (SELECT teamID, yearID, SUM(salary) AS total_spending
FROM salaries
GROUP BY teamID, yearID
ORDER BY teamID, yearID),
sp AS (SELECT teamID, AVG(total_spending) AS avg_spend,
NTILE(5) over(ORDER BY AVG(total_spending) DESC) AS spend_pct
FROM ts
GROUP BY teamID)
SELECT teamID, ROUND(avg_spend / 1000000, 1) AS avg_spend_millions
FROM sp
WHERE spend_pct = 1
--
```

teamID	avg_spend_millions
SFG	143.5
LAA	118.5
NYA	109.4
BOS	81.1
LAN	74.6
WAS	71.5
ARI	71.2
PHI	66.1

3. For each team, show the cumulative sum of spending over the years

```
WITH ts AS (SELECT teamID, yearID, SUM(salary) AS total_spend
            FROM salaries
            GROUP BY teamID, yearID
            ORDER BY teamID, yearID)
SELECT teamID, yearID,
        ROUND(SUM(total_spend) OVER(PARTITION BY teamID ORDER BY yearID) / 1000000, 1)
        AS Cumulative_Sum
FROM ts;
```

teamID	yearID	Cumulative_Sum
ANA	1997	31.1
ANA	1998	72.4
ANA	1999	127.8
ANA	2000	179.3
ANA	2001	226.8
ANA	2002	288.5
ANA	2003	367.6
ANA	2004	468.1
ARI	1998	32.3
ARI	1999	101.1
ARI	2000	182.1
ARI	2001	267.2
ARI	2002	370.0

4. Return the first year that each team's cumulative spending surpassed 1 billion

```sql
WITH ts AS (SELECT teamID, yearID, SUM(salary) AS total_spend
            FROM salaries
            GROUP BY teamID, yearID
            ORDER BY teamID, yearID),
    cs AS  (
            SELECT teamID, yearID,
            SUM(total_spend) OVER(PARTITION BY teamID ORDER BY yearID)
            AS Cumulative_Sum
            FROM ts),
    rn AS   (SELECT teamID, yearID, Cumulative_Sum,
            ROW_NUMBER() OVER(PARTITION BY teamID ORDER BY Cumulative_Sum) as rn
            FROM cs
            WHERE Cumulative_Sum > 1000000000)
SELECT teamID, yearID, ROUND(Cumulative_Sum / 1000000000, 2) AS Cumulative_Sum_Billions
FROM rn
WHERE rn = 1;
```

Result Grid | Filter Rows: | Export: | Wra

teamID	yearID	Cumulative_Sum_Billions
ARI	2012	1.02
ATL	2005	1.07
BAL	2007	1.06
BOS	2004	1.00
CHA	2008	1.07
CHN	2007	1.08
CIN	2010	1.06
CLE	2009	1.06
COL	2011	1.05
DET	2009	1.11
HOU	2008	1.03
KCA	2012	1.02
LAA	2013	1.06

PART III: PLAYER CAREER ANALYSIS

1. View the players table and find the number of players in the table

```
-- TASK 1. View the players table and find the number of players in the table
SELECT * FROM players;
SELECT COUNT(DISTINCT playerID) AS Total_players FROM players;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

playerID	birthYear	birthMonth	birthDay	birthCountry	birthState	birthCity	deathYear	deathMonth	deathDay	deathCountry
aardsda01	1981	12	27	USA	CO	Denver	NULL	NULL	NULL	NULL
aaronha01	1934	2	5	USA	AL	Mobile	NULL	NULL	NULL	NULL
aaronto01	1939	8	5	USA	AL	Mobile	1984	8	16	USA
aasedo01	1954	9	8	USA	CA	Orange	NULL	NULL	NULL	NULL
abadan01	1972	8	25	USA	FL	Palm Beach	NULL	NULL	NULL	NULL
abadfe01	1985	12	17	D.R.	La Romana	La Romana	NULL	NULL	NULL	NULL
abadijo01	1854	11	4	USA	PA	Philadelphia	1905	5	17	USA
abbated01	1877	4	15	USA	PA	Latrobe	1957	1	6	USA
abbeybe01	1869	11	11	USA	VT	Essex	1962	6	11	USA
abbeych01	1866	10	14	USA	NE	Falls City	1926	4	27	USA
abbotda01	1862	3	16	USA	OH	Portage	1930	2	13	USA
abbotfr01	1874	10	22	USA	OH	Versailles	1935	6	11	USA

Result Grid | Filter Rows:

Total_players
18589

-- 2. For each player, calculate their age at their first game, their last game, and their career length (all in years). Sort from longest career to shortest career.

```
SELECT  nameGiven,
        TIMESTAMPDIFF(YEAR, CAST(CONCAT(birthYear, '-', birthMonth, '-', birthDay) AS DATE),
        debut) AS starting_age,
        TIMESTAMPDIFF(YEAR, CAST(CONCAT(birthYear, '-', birthMonth, '-', birthDay) AS DATE),
        finalGame) AS end_age,
        TIMESTAMPDIFF(YEAR, debut, finalGame) AS career_length
FROM players
ORDER BY career_length DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch r

nameGiven	starting_age	end_age	career_length
Nicholas	21	57	35
James Henry	21	54	32
Saturnino Orestes Armas	23	54	31
Charles Timothy	28	58	30
Walter Arlington	20	49	29
Hugh Ambrose	22	49	27
James Thomas	20	48	27
Lynn Nolan	19	46	27
Charles Evard	21	48	27
John Joseph	21	48	27
Adrian Constantine	19	45	26
Dennis Joseph	21	46	25

-- 3. What team did each player play on for their starting and ending years?

```sql
-- TASK.3 What team did each player play on for their starting and ending years?
SELECT  p.nameGiven,
        s.yearID AS starting_year, s.teamID AS starting_team,
        e.yearID AS ending_year, e.teamID AS ending_team
FROM players p INNER JOIN salaries s
                    on p.playerID = s.playerID
                    AND YEAR(p.debut) = s.yearID
               INNER JOIN salaries e
                    on p.playerID = e.playerID
                    AND YEAR(P.finalGame) = e.yearID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

nameGiven	starting_year	starting_team	ending_year	ending_team
Kirk Edward	1985	CAL	1996	CHA
Mariano	1985	LAN	1997	NYA
Teodoro Valenzuela	1985	ML4	1994	ML4
Roger Alan	1985	NYN	1996	BAL
Timothy Dean	1985	OAK	1990	CIN
Vincent Maurice	1985	SLN	1997	DET
Paul Andre	1986	ATL	1999	CLE
Edward R.	1986	ATL	1988	ATL
Robert Clifford	1986	ATL	1986	ATL
Charles Edward	1986	CAL	2002	CLE
Wallace Keith	1986	CAL	2001	ANA
Roberto Martin An...	1986	CHA	2001	SLN
Ronald Joseph	1986	CHA	1997	CHA

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-- 4. How many players started and ended on the same team and also played for over a decade?

```sql
-- 4. How many players started and ended on the same team
and also played for over a decade                         */
SELECT  p.nameGiven,
        s.yearID AS starting_year, s.teamID AS starting_team,
        e.yearID AS ending_year, e.teamID AS ending_team
FROM players p INNER JOIN salaries s
                    on p.playerID = s.playerID
                    AND YEAR(p.debut) = s.yearID
               INNER JOIN salaries e
                    on p.playerID = e.playerID
                    AND YEAR(P.finalGame) = e.yearID
WHERE s.teamID = e.teamID
AND e.yearID - s.yearID > 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

nameGiven	starting_year	starting_team	ending_year	ending_team
Ronald Joseph	1986	CHA	1997	CHA
Barry Louis	1986	CIN	2004	CIN
Thomas Michael	1987	ATL	2008	ATL
Ellis Rena	1987	BOS	2004	BOS
Thomas Alan	1987	SLN	1998	SLN
George Kenneth	1989	SEA	2010	SEA
Samuel Peralta	1989	TEX	2007	TEX
David Michael	1990	PHI	2002	PHI
Raymond Lewis	1990	SLN	2004	SLN
Bernabe	1991	NYA	2006	NYA
Patrick George	1991	TOR	2004	TOR
Larry Wayne	1993	ATL	2012	ATL
Brad William	1995	MIN	2006	MIN

PART IV: PLAYER COMPARISON ANALYSIS

-- 1. View the players table

```
1    -- 1. View the players table
2 •  SELECT * FROM players;
```

	playerID	birthYear	birthMonth	birthDay	birthCountry	birthState	birthCity	deathYear	deathMonth	deathDay	deathCountry
▸	aardsda01	1981	12	27	USA	CO	Denver	NULL	NULL	NULL	NULL
	aaronha01	1934	2	5	USA	AL	Mobile	NULL	NULL	NULL	NULL
	aaronto01	1939	8	5	USA	AL	Mobile	1984	8	16	USA
	aasedo01	1954	9	8	USA	CA	Orange	NULL	NULL	NULL	NULL
	abadan01	1972	8	25	USA	FL	Palm Beach	NULL	NULL	NULL	NULL
	abadfe01	1985	12	17	D.R.	La Romana	La Romana	NULL	NULL	NULL	NULL
	abadijo01	1854	11	4	USA	PA	Philadelphia	1905	5	17	USA
	abbated01	1877	4	15	USA	PA	Latrobe	1957	1	6	USA

players 1

-- 2. Which players have the same birthday?

```
WITH bn AS (SELECT  CAST(concat(birthYear, '-', birthMonth, '-', birthDay) AS DATE)
        AS birthdate,nameGiven
        FROM players)
SELECT birthdate, GROUP_CONCAT(nameGiven SEPARATOR ', ') AS players
FROM bn
WHERE YEAR(birthdate) BETWEEN 1980 AND 1990
GROUP BY birthdate
ORDER BY birthdate;
```

	birthdate	players
▸	1980-01-03	Bradley Keith
	1980-01-10	Matthew Stephen
	1980-01-12	Robert Edward
	1980-01-15	Jeffrey Darrin, Matthew Thomas
	1980-01-16	Brooks Litchfield, Jose Alberto
	1980-01-17	Thomas Joseph, Michael Gregory
	1980-01-20	Franklyn Miguel, Luis
	1980-01-25	Phillip Matthew
	1980-01-26	Brandon Edward, Antonio Miguel
	1980-02-01	Hector R.
	1980-02-03	Jared Michael
	1980-02-04	Stephen John, Douglas
	1980-02-07	Brad Martin

Result 12

-- 3. Create a summary table that shows for each team, what percent of players bat right, left and both

```sql
SELECT teamID,
        ROUND(SUM(CASE WHEN bats = 'R' THEN 1 ELSE 0 END) / COUNT(playerID) * 100, 1)
        AS bats_right,
        ROUND(SUM(CASE WHEN bats = 'L' THEN 1 ELSE 0 END) / COUNT(playerID) * 100, 1)
        AS bats_left,
        ROUND(SUM(CASE WHEN bats = 'B' THEN 1 ELSE 0 END) / COUNT(playerID) * 100, 1)
        AS bats_both
FROM up
GROUP BY teamID;
```

Result Grid | Filter Rows: | Export: | Wr

teamID	bats_right	bats_left	bats_both
ATL	62.0	30.6	7.4
BAL	64.4	27.0	8.6
BOS	63.1	27.8	8.8
CAL	61.9	28.8	9.4
CHA	60.2	31.8	8.0
CHN	59.4	32.1	8.5
CIN	60.4	29.9	9.6
CLE	61.2	29.6	9.2
DET	62.0	27.4	10.5
HOU	59.3	28.6	12.0
KCA	63.7	27.7	8.6
LAN	62.2	28.2	9.3
MIN	63.0	24.1	12.9

-- 4. How have average height and weight at debut game changed over the years, and what's the decade-over-decade difference?

```
WITH hw AS (SELECT  FLOOR(YEAR(debut) / 10) * 10 AS decade,
                    AVG(height) AS avg_height, AVG(weight) AS avg_weight
            FROM    players
            GROUP BY decade)

SELECT  decade,
        avg_height - LAG(avg_height) OVER(ORDER BY decade) AS height_diff,
        avg_weight - LAG(avg_weight) OVER(ORDER BY decade) AS weight_diff
FROM    hw
WHERE   decade IS NOT NULL;
```

Result Grid | Filter Rows: | Expo

decade	height_diff	weight_diff
1870	NULL	NULL
1880	0.7423	5.8693
1890	0.4023	1.3236
1900	0.5436	3.7460
1910	0.2519	-2.2125
1920	0.1276	1.2309
1930	0.7343	5.7174
1940	0.4079	3.5361
1950	0.4140	2.0629
1960	0.4139	1.4574
1970	0.1921	0.1835
1980	0.2722	1.6483
1990	0.1460	6.1865

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